# Math 54 — Week 5

## Properties of determinants; Cramer's rule and volume; vector spaces; null and column spaces (Lay 3.2–3.3, 4.1–4.2)

## §1 Facts used from Weeks 1–4

(W1) (Week 4) $\det A$ can be computed by cofactor expansion along any row or down any column; for a triangular matrix it is the product of the diagonal entries.

(W2) (Week 4) An elementary matrix $E$ performs a row operation: $EA$ is $A$ after that operation. The inverse of an elementary matrix is elementary. If $A$ is invertible, then $E_k \cdots E_1 A = I$ for some elementary $E_i$, so $A = E_1^{-1} \cdots E_k^{-1}$ is a product of elementary matrices.

(W3) (Week 4) Invertible matrix theorem: for square $A$, invertible $\iff$ $n$ pivots $\iff$ $A\mathbf{x} = \mathbf{0}$ has only the trivial solution $\iff$ the columns are independent $\iff$ the columns span $\mathbb{R}^n$ $\iff$ $A^T$ is invertible. If $A, B$ are square and $AB$ is invertible, so are $A$ and $B$. Column $j$ of $A^{-1}$ solves $A\mathbf{x} = \mathbf{e}_j$. $(AB)^T = B^T A^T$.

(W4) (Week 2) The solution set of $A\mathbf{x} = \mathbf{0}$ is the span of the vectors in its parametric vector form. $A\mathbf{x} = \mathbf{b}$ is consistent exactly when $\mathbf{b}$ is a combination of the columns of $A$, and this holds for every $\mathbf{b}$ exactly when $A$ has a pivot in every row.

(W5) (Weeks 2–3) In an echelon form, each nonzero row has a pivot and the zero rows are at the bottom. In an $n \times n$ echelon form with $n$ pivots, the pivot of row $i$ is in column $i$ (the argument of Week 4 for the reduced form uses only the echelon conditions).

## §2 This week

### §2.1 Properties of determinants (Lay 3.2)

**The question.** Expanding an $n \times n$ determinant by cofactors needs $n$ determinants of size $n-1$, so at least $n!$ multiplications: $10! = 3\,628\,800$ for a $10 \times 10$ matrix. Row reduction of a $10 \times 10$ matrix to echelon form takes about $n^3/3 \approx 330$ multiplications: clearing the entries below pivot $k$ takes about $(n-k)^2$ of them, and $\sum_k (n-k)^2 \approx n^3/3$. How does each row operation change the determinant, so that row reduction can compute it? For example, what is $\det A$ for the matrix below?

$$A = \begin{bmatrix} 1 & -4 & 2 \\ -2 & 8 & -9 \\ -1 & 7 & 0 \end{bmatrix}$$

**Theorem** (Row operations (Lay 3.2, Thm. 3))

Let $A$ be a square matrix.

(a) If a multiple of one row of $A$ is added to another row to produce $B$, then $\det B = \det A$.

(b) If two rows of $A$ are interchanged to produce $B$, then $\det B = -\det A$.

(c) If one row of $A$ is multiplied by $k$ to produce $B$, then $\det B = k \det A$.

**The idea**

A cofactor expansion along a row that the operation did not change uses the same entries and signs for $A$ and $B$, and minors that differ by the same kind of operation one size smaller. So each statement for size $n$ follows from the statement for size $n-1$, starting from a direct $2 \times 2$ computation.

*Proof.* 1. (c) Expand $\det B$ along the scaled row $i$. Its entries are $ka_{ij}$, and its cofactors are those of $A$, because deleting row $i$ removes the only change. So $\det B = \sum_j ka_{ij}C_{ij} = k \det A$.

2. (a), (b) for $n = 2$. Interchanging the rows, and adding $k$ times row 1 to row 2:

$$\det \begin{bmatrix} c & d \\ a & b \end{bmatrix} = cb - da = -(ad - bc), \qquad \det \begin{bmatrix} a & b \\ c + ka & d + kb \end{bmatrix} = a(d + kb) - b(c + ka) = ad - bc.$$

Adding $k$ times row 2 to row 1:

$$\det \begin{bmatrix} a + kc & b + kd \\ c & d \end{bmatrix} = (a + kc)d - (b + kd)c = ad - bc.$$

3. (a), (b) for $n \geq 3$, **assuming them for size** $n - 1$. The operation involves at most two rows, so some row $i$ is unchanged. Expanding along row $i$,

$$\det B = \sum_j a_{ij}(-1)^{i+j} \det B_{ij},$$

and each $B_{ij}$ is $A_{ij}$ after the same kind of operation. By the case $n - 1$, $\det B_{ij} = \det A_{ij}$ for a replacement and $\det B_{ij} = -\det A_{ij}$ for an interchange. Summing over $j$ gives $\det B = \det A$, respectively $\det B = -\det A$.

□

**Determinant by row reduction.** Reduce $A$ to an echelon form $U$ using only replacements and $r$ interchanges. Each replacement keeps the determinant and each interchange changes its sign, so $\det A = (-1)^r \det U$; $U$ is square and in echelon form, hence triangular, so by (W1)

$$\det A = (-1)^r \cdot (\text{product of the diagonal entries of } U).$$

**The question answered.**

$$\begin{bmatrix} 1 & -4 & 2 \\ -2 & 8 & -9 \\ -1 & 7 & 0 \end{bmatrix} \xrightarrow[R_3 + R_1]{R_2 + 2R_1} \begin{bmatrix} 1 & -4 & 2 \\ 0 & 0 & -5 \\ 0 & 3 & 2 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{bmatrix} 1 & -4 & 2 \\ 0 & 3 & 2 \\ 0 & 0 & -5 \end{bmatrix}.$$

One interchange, so $\det A = (-1)^1 \cdot (1)(3)(-5) = 15$.

**Two equal rows, or a zero row, give determinant** 0. If rows $i$ and $j$ are equal, the replacement $R_i - R_j$ keeps the determinant and produces a zero row. Expanding along a zero row gives 0.

**Theorem** (Invertibility (Lay 3.2, Thm. 4))

A square matrix $A$ is invertible if and only if $\det A \neq 0$.

*Proof.* 1. Reduce $A$ to an echelon form $U$ using only replacements and interchanges. By Theorem 3(a), (b), $\det A = \pm \det U$, so $\det A \neq 0$ exactly when $\det U \neq 0$.

2. **Assume** $A$ **has** $n$ **pivots.** By (W5) the pivots of $U$ are its diagonal entries, so $\det U$ is the product of the pivots, which is not 0.

3. **Assume $A$ has fewer than $n$ pivots.** Then $U$ has a row without a pivot, which is a zero row and lies at the bottom (W5). So the last diagonal entry of $U$ is 0 and $\det U = 0$.
4. So $\det A \neq 0$ exactly when $A$ has $n$ pivots, which by the invertible matrix theorem (W3) is exactly when $A$ is invertible.

□

This adds a statement to the invertible matrix theorem. In particular, **the columns of a square matrix are linearly independent if and only if** $\det A \neq 0$.

**Theorem** (Transpose (Lay 3.2, Thm. 5))
If $A$ is $n \times n$, then $\det A^T = \det A$.

*Proof.* 1. By induction on $n$. For $n = 1$, $A^T = A$.

2. **Assume the statement for size $n-1$.** Deleting row 1 and column $j$ of $A^T$ gives $(A_{j1})^T$, so the expansion of $\det A^T$ along row 1 is $\sum_j a_{j1}(-1)^{1+j} \det(A_{j1})^T$.
3. By the assumption, $\det(A_{j1})^T = \det A_{j1}$, so this sum is the expansion of $\det A$ down column 1, which equals $\det A$ (W1).

□

Rows of $A^T$ are columns of $A$, so **column operations change the determinant exactly as row operations do**: a column replacement keeps it, a column interchange changes its sign, and scaling a column by $k$ multiplies it by $k$.

**Theorem** (Multiplicative property (Lay 3.2, Thm. 6))
If $A$ and $B$ are $n \times n$, then $\det(AB) = (\det A)(\det B)$.

*Proof.* 1. **For an elementary $E$ and any $M$, $\det(EM) = (\det E)(\det M)$.** $EM$ is $M$ after one row operation (W2), so by Theorem 3 $\det(EM)$ is $\det M$, $-\det M$ or $k \det M$. The same theorem applied to $M = I$ gives $\det E = 1$, $-1$ or $k$ for the same operation.

2. **Assume $A$ is not invertible.** Then $AB$ is not invertible (W3), so by Theorem 4 both $\det(AB)$ and $(\det A)(\det B)$ are 0.
3. **Assume $A$ is invertible.** Then $A = E_p \cdots E_1$ for elementary matrices (W2). Applying step 1 $p$ times,

$$\det(AB) = \det(E_p \cdots E_1 B) = (\det E_p) \cdots (\det E_1) \det B, \qquad \det A = (\det E_p) \cdots (\det E_1).$$

So $\det(AB) = (\det A)(\det B)$.

□

**Consequences.** For $n \times n$ matrices:

- If $A$ is invertible, $(\det A)(\det A^{-1}) = \det(AA^{-1}) = \det I = 1$, so

$$\det(A^{-1}) = \frac{1}{\det A}.$$

- $\det(A^k) = (\det A)^k$, by Theorem 6 applied $k-1$ times.
- $\det(cA) = c^n \det A$, because $cA = (cI)A$ and $\det(cI) = c^n$ ($cI$ is diagonal).
- **Linearity in one column**: with all other columns fixed, $\mathbf{x} \mapsto \det[\,\mathbf{a}_1 \cdots \mathbf{x} \cdots \mathbf{a}_n\,]$ is a linear function of $\mathbf{x}$. Reason: expanding down that column gives $x_1 C_{1j} + \cdots + x_n C_{nj}$, and the cofactors $C_{ij}$ do not involve $\mathbf{x}$.

**Shortcuts**

- **Determinant by row reduction**: use only replacements and interchanges, count the interchanges, multiply the diagonal. Reason: Theorem 3. Instance: the question.

- **Pull out a common factor** of a row (or column) first; it multiplies the determinant by that factor. Reason: Theorem 3(c). Instance: row 1 of P1 is $2(1, 2, -1, 3)$, so det of P1's matrix is 2 times the determinant with row 1 replaced by $(1, 2, -1, 3)$.
- **Combine expansion and reduction**: create zeros in one column with replacements (which keep det), then expand down that column. Reason: Theorem 3(a) and (W1). Instance: $R_2 - 2R_1$ and $R_3 - 3R_1$ give

$$\det \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 3 & 5 & 9 \end{bmatrix} = \det \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix} = 1 \cdot \det \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} = 1,$$

  expanding down column 1.
- **Invertible for which $h$?** Compute det as a polynomial in $h$; the matrix is invertible exactly away from its zeros. Reason: Theorem 4. Instance: the matrix below has $\det = 4 - 2h$, so it is invertible exactly when $h \neq 2$.

$$\begin{bmatrix} 1 & h \\ 2 & 4 \end{bmatrix}$$

- **Determinant rules**: $\det(AB) = \det A \det B$, $\det A^{-1} = 1/\det A$, $\det A^T = \det A$, $\det(cA) = c^n \det A$. Instance: for $3 \times 3$ $A$ with $\det A = 5$, $\det(2A^{-1}) = 2^3 \cdot \frac{1}{5} = \frac{8}{5}$.

**Traps**

- Scaling a row by $k$ during row reduction multiplies det by $k$; divide it back out. Scaling row 1 of the matrix below on the left by $\frac{1}{2}$ gives the matrix on the right, with determinant 3, while the original determinant is $2 \cdot 3 = 6$.

$$\begin{bmatrix} 2 & 4 \\ 0 & 3 \end{bmatrix} \qquad \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$$

- $\det(cA) = c^n \det A$, not $c \det A$: $\det(2I_2) = 4$. And $\det(A + B) \neq \det A + \det B$ in general: for $2 \times 2$, $\det(I + (-I)) = 0$ but $\det I + \det(-I) = 2$.
- $\det A = 0$ does not require two equal rows or a zero row. The matrix below has neither, and no two of its rows are multiples of each other, yet row 3 = 2(row 2) − row 1, so det = 0.

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

- "Product of the diagonal" is the determinant only for triangular matrices: the matrix below has diagonal product 0 and determinant $-1$.

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

## §2.2 Cramer's rule, the inverse formula, and volume (Lay 3.3)

**The question.** Can the system $2x_1 + x_2 = 7$, $-3x_1 + 4x_2 = 6$ be solved with determinants alone, without row reduction? And what is the area of the parallelogram with sides $(2, -3)$ and $(1, 4)$?

**Notation.** For an $n \times n$ matrix $A$ and $\mathbf{b}$ in $\mathbb{R}^n$, $A_i(\mathbf{b})$ is $A$ with column $i$ replaced by $\mathbf{b}$.

**Theorem** (Cramer's rule (Lay 3.3, Thm. 7))

If $A$ is an invertible $n \times n$ matrix, then for each $\mathbf{b}$ the unique solution of $A\mathbf{x} = \mathbf{b}$ has entries

$$x_i = \frac{\det A_i(\mathbf{b})}{\det A}, \qquad i = 1, \ldots, n.$$

**The idea**

The identity matrix with column $i$ replaced by $\mathbf{x}$ has determinant $x_i$, and multiplying it by $A$ gives $A$ with column $i$ replaced by $A\mathbf{x} = \mathbf{b}$. Taking determinants of that product gives the formula.

*Proof.* 1. Let $I_i(\mathbf{x})$ be $I$ with column $i$ replaced by the solution $\mathbf{x}$. Column $j \neq i$ of $AI_i(\mathbf{x})$ is $A\mathbf{e}_j = \mathbf{a}_j$, and column $i$ is $A\mathbf{x} = \mathbf{b}$, so $AI_i(\mathbf{x}) = A_i(\mathbf{b})$.

2. Expand $\det I_i(\mathbf{x})$ along row $i$: its only nonzero entry is $x_i$ in position $(i, i)$, with sign $+$, and deleting row $i$ and column $i$ leaves $I_{n-1}$. So $\det I_i(\mathbf{x}) = x_i$.

3. By Theorem 6, $(\det A)\, x_i = \det A_i(\mathbf{b})$. Divide by $\det A \neq 0$ (Theorem 4).

□

**The question answered (system).**

$$\det A = \det \begin{bmatrix} 2 & 1 \\ -3 & 4 \end{bmatrix} = 11, \quad \det A_1(\mathbf{b}) = \det \begin{bmatrix} 7 & 1 \\ 6 & 4 \end{bmatrix} = 22, \quad \det A_2(\mathbf{b}) = \det \begin{bmatrix} 2 & 7 \\ -3 & 6 \end{bmatrix} = 33,$$

so $x_1 = 22/11 = 2$ and $x_2 = 33/11 = 3$. Check: $4 + 3 = 7$ and $-6 + 12 = 6$.

**Worked example: Cramer's rule with a parameter.** For which $s$ does $sx_1 + 2x_2 = 1$, $3x_1 + sx_2 = 2$ have a unique solution, and what is it?

1. $\det A = s^2 - 6$, so the solution is unique exactly when $s \neq \pm\sqrt{6}$ (Theorem 4).

2. For those $s$,
$$x_1 = \frac{1}{s^2 - 6} \det \begin{bmatrix} 1 & 2 \\ 2 & s \end{bmatrix} = \frac{s - 4}{s^2 - 6}, \qquad x_2 = \frac{1}{s^2 - 6} \det \begin{bmatrix} s & 1 \\ 3 & 2 \end{bmatrix} = \frac{2s - 3}{s^2 - 6}.$$

3. Check with $s = 0$: the system is $2x_2 = 1$, $3x_1 = 2$, so $x_1 = \frac{2}{3}$ and $x_2 = \frac{1}{2}$; the formulas give $\frac{-4}{-6}$ and $\frac{-3}{-6}$.

**Theorem** (An inverse formula (Lay 3.3, Thm. 8))

If $A$ is invertible, then

$$A^{-1} = \frac{1}{\det A} \operatorname{adj} A,$$

where the **adjugate** $\operatorname{adj} A$ is the *transpose* of the matrix of cofactors: $(\operatorname{adj} A)_{ij} = C_{ji}$.

*Proof.* 1. Column $j$ of $A^{-1}$ is the solution of $A\mathbf{x} = \mathbf{e}_j$ (W3). By Cramer's rule, its entry $i$ is $\det A_i(\mathbf{e}_j)/\det A$.

2. Column $i$ of $A_i(\mathbf{e}_j)$ is $\mathbf{e}_j$, with a single 1 in row $j$. Expanding down column $i$ gives $(-1)^{j+i}$ times the determinant of $A_i(\mathbf{e}_j)$ without row $j$ and column $i$, which is $A$ without row $j$ and column $i$. So $\det A_i(\mathbf{e}_j) = (-1)^{i+j} \det A_{ji} = C_{ji}$.

□

**Worked example: a $3 \times 3$ adjugate.** For the matrix below, expansion along row 1 gives $\det A = 1 \cdot 1 - 2 \cdot (0 - 1) + 0 = 3$. The nine cofactors are:

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}, \qquad \begin{aligned} C_{11} &= +\det \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = 1, & C_{12} &= -\det \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} = 1, & C_{13} &= +\det \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = -1, \\ C_{21} &= -\det \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} = -2, & C_{22} &= +\det \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = 1, & C_{23} &= -\det \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix} = 2, \\ C_{31} &= +\det \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix} = 2, & C_{32} &= -\det \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = -1, & C_{33} &= +\det \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} = 1. \end{aligned}$$

The adjugate has the cofactors $C_{1j}$ as its *column* 1, and so on:

$$\operatorname{adj} A = \begin{bmatrix} 1 & -2 & 2 \\ 1 & 1 & -1 \\ -1 & 2 & 1 \end{bmatrix}, \qquad A^{-1} = \frac{1}{3} \begin{bmatrix} 1 & -2 & 2 \\ 1 & 1 & -1 \\ -1 & 2 & 1 \end{bmatrix}.$$

Check one column: row 1 of $A$ times column 1 of $\operatorname{adj} A$ is $1 + 2 + 0 = 3$, and row 2 times column 1 is $0 + 1 - 1 = 0$, as $A \operatorname{adj} A = 3I$ requires. For numerical work, row reducing $[\,A \;\; I\,]$ is faster; the formula is for $2 \times 2$ matrices (it gives the formula of Week 4) and for theory.

**Determinants as area and volume.**

**Theorem** (Area and volume (Lay 3.3, Thm. 9))

If $A$ is $2 \times 2$, the area of the parallelogram determined by the columns of $A$ is $|\det A|$. If $A$ is $3 \times 3$, the volume of the parallelepiped determined by the columns of $A$ is $|\det A|$.

**The idea**

A column replacement $\mathbf{a}_2 \to \mathbf{a}_2 + c\mathbf{a}_1$ slides the far side of the parallelogram along the line of $\mathbf{a}_1$: the base and the height stay the same, so the area does, and the determinant does too. Such moves turn any parallelogram into a rectangle with sides along the axes, where area and $|\det|$ are both the product of the side lengths.

*Proof for* $2 \times 2$. 1. **Assume the columns are dependent.** Then they lie on one line through $\mathbf{0}$, the parallelogram is flat with area 0, and $\det A = 0$ (Theorem 4).

2. **Assume the columns are independent**, so $\det A \neq 0$. A column replacement $\mathbf{a}_2 \to \mathbf{a}_2 + c\mathbf{a}_1$ keeps det (column version of Theorem 3(a)) and keeps the area: the new parallelogram has the same base $\mathbf{a}_1$, and $\mathbf{a}_2 + c\mathbf{a}_1$ differs from $\mathbf{a}_2$ by a vector along $\mathbf{a}_1$, so it is the same distance from the line of $\mathbf{a}_1$ (same height). The same holds for $\mathbf{a}_1 \to \mathbf{a}_1 + c\mathbf{a}_2$. A column interchange keeps $|\det|$ and the parallelogram.

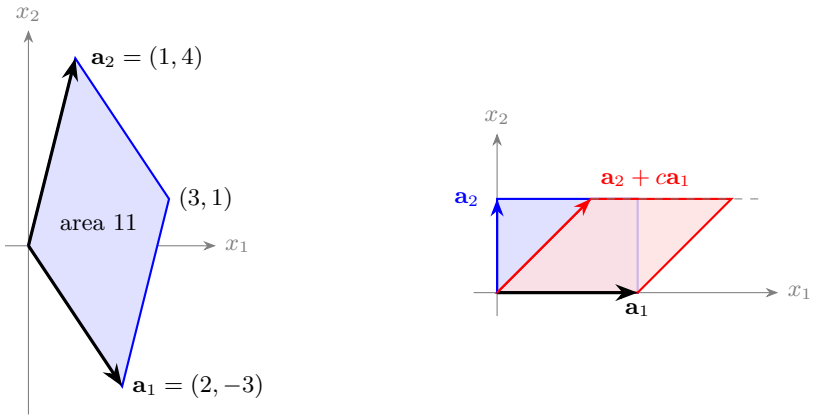
3. Column operations on $A$ are row operations on $A^T$. Reduce $A^T$ to echelon form by replacements and interchanges; it has two pivots $p, r \neq 0$, so it is upper triangular, and $A$ has become its transpose, which is lower triangular:

$$A^T \to \begin{bmatrix} p & q \\ 0 & r \end{bmatrix}, \qquad A \to \begin{bmatrix} p & 0 \\ q & r \end{bmatrix}.$$

   One more column replacement, column 1 minus $\frac{q}{r}$ times column 2, gives $\operatorname{diag}(p, r)$.

4. The parallelogram of $\operatorname{diag}(p, r)$ is a rectangle with sides $|p|$ and $|r|$, of area $|pr| = |\det \operatorname{diag}(p, r)|$. All the steps kept both the area and $|\det|$, so the area of the original parallelogram is $|\det A|$.

The $3 \times 3$ case is proved the same way with parallelepipeds (base area times height); it is cited here. □




Left: the question's parallelogram, with sides $\mathbf{a}_1$, $\mathbf{a}_2$ and fourth vertex $\mathbf{a}_1 + \mathbf{a}_2 = (3,1)$. Right: a column replacement keeps the base $\mathbf{a}_1$ and the height (the top side stays on the dashed line), so it keeps the area.

**The question answered (area).** The parallelogram has area $|\det A| = |2 \cdot 4 - 1 \cdot (-3)| = 11$.

**Parallelograms not at the origin.** The parallelogram with vertices $\mathbf{p}$, $\mathbf{p}+\mathbf{u}$, $\mathbf{p}+\mathbf{v}$, $\mathbf{p}+\mathbf{u}+\mathbf{v}$ is the one with sides $\mathbf{u}, \mathbf{v}$ moved by $\mathbf{p}$. Moving a region without turning it does not change its area, so its area is $|\det[\,\mathbf{u}\ \ \mathbf{v}\,]|$.

**Theorem** (Linear maps scale area and volume (Lay 3.3, Thm. 10))
Let $T\colon \mathbb{R}^2 \to \mathbb{R}^2$ be $T(\mathbf{x}) = A\mathbf{x}$. If $S$ is a parallelogram, then $\text{area}(T(S)) = |\det A| \cdot \text{area}(S)$. The same holds for every region $S$ with finite area, and in $\mathbb{R}^3$ with volume in place of area.

*Proof for parallelograms.* 1. Let $S$ have sides $\mathbf{b}_1, \mathbf{b}_2$ at $\mathbf{0}$: $S = \{s_1\mathbf{b}_1 + s_2\mathbf{b}_2 : 0 \le s_1, s_2 \le 1\}$. By linearity, $T(S) = \{s_1 A\mathbf{b}_1 + s_2 A\mathbf{b}_2\}$, the parallelogram with sides $A\mathbf{b}_1$ and $A\mathbf{b}_2$.

2. With $B = [\,\mathbf{b}_1\ \ \mathbf{b}_2\,]$, Theorem 9 and Theorem 6 give

$$\text{area}(T(S)) = |\det[\,A\mathbf{b}_1\ \ A\mathbf{b}_2\,]| = |\det(AB)| = |\det A|\,|\det B| = |\det A| \cdot \text{area}(S).$$

3. A parallelogram $\mathbf{p} + S$ goes to $T(\mathbf{p}) + T(S)$, a moved copy of $T(S)$, with the same area.

For general regions, Lay approximates the region by small squares; that step is cited here. □

**Worked example: area of an ellipse.** The ellipse $x_1^2/a^2 + x_2^2/b^2 \le 1$ $(a, b > 0)$ is the image of the unit disk under $\mathbf{x} \mapsto \text{diag}(a,b)\mathbf{x}$, because $(u_1, u_2)$ with $u_1^2 + u_2^2 \le 1$ goes to $(au_1, bu_2)$. So its area is $|ab| \cdot \pi = \pi ab$.

### Shortcuts

- **Cramer's rule** is fastest for a $2 \times 2$ system, or for one entry $x_i$ of a larger system. Reason: each entry costs one extra determinant. Instance: the question.
- **Area of a triangle** with vertices $\mathbf{0}, \mathbf{u}, \mathbf{v}$: half the parallelogram, $\frac{1}{2}|\det[\,\mathbf{u}\ \mathbf{v}\,]|$. Instance: $\mathbf{0}$, $(3,0)$, $(1,2)$ give $\frac{1}{2}|6 - 0| = 3$.
- **Triangle with vertices** $(x_i, y_i)$: the area is half the absolute value of

$$\det \begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix}$$

  Reason: $R_2 - R_1$ and $R_3 - R_1$ keep the determinant and make column 3 equal to $(1,0,0)$; expanding down column 3 leaves the $2 \times 2$ determinant below, which by Theorem 5 is $\det[\,\mathbf{u}\ \mathbf{v}\,]$ for the sides $\mathbf{u}, \mathbf{v}$ from vertex 1.

$$\det \begin{bmatrix} x_2 - x_1 & y_2 - y_1 \\ x_3 - x_1 & y_3 - y_1 \end{bmatrix}$$

  Instance: $(1,1)$, $(4,1)$, $(2,3)$ give $\frac{1}{2}|3 \cdot 2 - 0 \cdot 1| = 3$.

- **Area of an image**: multiply the area by $|\det A|$. Reason: Theorem 10. Instance: the ellipse.

**Traps**

- The adjugate is the *transpose* of the cofactor matrix. For the matrix below, the cofactors are $C_{11} = 4$, $C_{12} = -3$, $C_{21} = -2$, $C_{22} = 1$, so the adjugate is the transposed arrangement:

$$\operatorname{adj}\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix} = \begin{bmatrix}4 & -2\\ -3 & 1\end{bmatrix}, \qquad \text{not} \qquad \begin{bmatrix}4 & -3\\ -2 & 1\end{bmatrix}.$$

- Area is $|\det|$; the sign of det records orientation. The columns $(0,1)$ and $(1,0)$ give det $= -1$ and area 1.
- Cramer's rule needs $\det A \neq 0$. In the parameter example with $s = \sqrt{6}$ the formulas divide by 0; then the system has no solution or infinitely many.

## §2.3 Vector spaces and subspaces (Lay 4.1)

**The question.** Polynomials, functions and matrices can all be added and scaled like vectors in $\mathbb{R}^n$. Among the polynomials of degree at most 2, do those with $p(1) = 0$ form a "space" in which sums and multiples stay inside, like a plane through the origin? Do those with $p(1) = 1$? What exactly is needed for span and linear combination to work, and how do we test a set?

**Definition.** A **vector space** is a nonempty set $V$ of objects (**vectors**) with an addition and a multiplication by real scalars such that, for all $\mathbf{u}, \mathbf{v}, \mathbf{w}$ in $V$ and all scalars $c, d$:

(1) $\mathbf{u} + \mathbf{v}$ is in $V$;
(2) $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$;
(3) $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$;
(4) there is a $\mathbf{0}$ in $V$ with $\mathbf{u} + \mathbf{0} = \mathbf{u}$;
(5) each $\mathbf{u}$ has a $-\mathbf{u}$ in $V$ with $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$;
(6) $c\mathbf{u}$ is in $V$;
(7) $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$;
(8) $(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$;
(9) $c(d\mathbf{u}) = (cd)\mathbf{u}$;
(10) $1\mathbf{u} = \mathbf{u}$.

**Examples.** In each, the operations are the familiar ones and the axioms reduce to rules for numbers.

- $\mathbb{R}^n$, with zero vector $(0, \ldots, 0)$.
- $\mathbb{P}_n$, the polynomials $a_0 + a_1 t + \cdots + a_n t^n$ of degree at most $n$, added and scaled coefficient by coefficient; the zero vector is the zero polynomial, all of whose coefficients are 0. $\mathbb{P}$ is the space of all polynomials.
- The real-valued functions on a set $D$, with $(f + g)(t) = f(t) + g(t)$ and $(cf)(t) = cf(t)$; the zero vector is the zero function, with value 0 at every $t$. The continuous functions on $[a, b]$ form the space $C[a, b]$.
- The $m \times n$ matrices, with the zero matrix as zero vector.

**Theorem** (Consequences of the axioms (Lay 4.1))
In every vector space, for every $\mathbf{u}$ and scalar $c$: $0\mathbf{u} = \mathbf{0}$, $c\mathbf{0} = \mathbf{0}$, and $-\mathbf{u} = (-1)\mathbf{u}$.

*Proof.* 1. By (8), $0\mathbf{u} = (0 + 0)\mathbf{u} = 0\mathbf{u} + 0\mathbf{u}$. Add $-(0\mathbf{u})$ to both sides and use (3), (5), (4): $\mathbf{0} = 0\mathbf{u}$.

2. By (4) and (7), $c\mathbf{0} = c(\mathbf{0} + \mathbf{0}) = c\mathbf{0} + c\mathbf{0}$. Add $-(c\mathbf{0})$ to both sides as in step 1: $\mathbf{0} = c\mathbf{0}$.

3. A vector has only one negative: if $\mathbf{u} + \mathbf{w} = \mathbf{0}$ and $\mathbf{u} + \mathbf{w}' = \mathbf{0}$, then $\mathbf{w} = \mathbf{w} + (\mathbf{u} + \mathbf{w}') = (\mathbf{w} + \mathbf{u}) + \mathbf{w}' = \mathbf{w}'$, by (4), (3), (2).

4. By (10), (8) and step 1, $\mathbf{u} + (-1)\mathbf{u} = 1\mathbf{u} + (-1)\mathbf{u} = (1 + (-1))\mathbf{u} = 0\mathbf{u} = \mathbf{0}$. So $(-1)\mathbf{u}$ is the negative of $\mathbf{u}$.

□

**Definition.** A **subspace** of a vector space $V$ is a subset $H$ of $V$ with three properties: (a) $\mathbf{0}$ is in $H$; (b) $H$ is closed under addition: $\mathbf{u}+\mathbf{v}$ is in $H$ whenever $\mathbf{u},\mathbf{v}$ are; (c) $H$ is closed under scalar multiplication: $c\mathbf{u}$ is in $H$ whenever $\mathbf{u}$ is in $H$ and $c$ is a scalar.

**A subspace is a vector space** with the operations of $V$: axioms (1), (6) are (b), (c); (4) holds by (a); (5) holds because $-\mathbf{u}=(-1)\mathbf{u}$ is in $H$ by (c) and the consequences of the axioms (Lay 4.1); the other axioms are equations that hold for all vectors of $V$, so in particular for those of $H$.

- $\{\mathbf{0}\}$ and $V$ are subspaces of $V$. $\mathbb{P}_n$ is a subspace of $\mathbb{P}$.
- A line through the origin in $\mathbb{R}^2$ is a subspace; a line not through the origin is not (it misses $\mathbf{0}$).
- $\mathbb{R}^2$ is *not* a subset, hence not a subspace, of $\mathbb{R}^3$: its vectors have two entries, not three. The plane $\{(s,t,0)\}$ in $\mathbb{R}^3$ is a subspace.

**Theorem** (A span is a subspace (Lay 4.1, Thm. 1))
If $\mathbf{v}_1,\dots,\mathbf{v}_p$ are in a vector space $V$, then $\mathrm{Span}\{\mathbf{v}_1,\dots,\mathbf{v}_p\}$ is a subspace of $V$.

*Proof.* 1. $\mathbf{0}=0\mathbf{v}_1+\cdots+0\mathbf{v}_p$ (by the consequences of the axioms), so $\mathbf{0}$ is in the span.

2. By the axioms, $\sum c_i\mathbf{v}_i+\sum d_i\mathbf{v}_i=\sum(c_i+d_i)\mathbf{v}_i$ and $k\sum c_i\mathbf{v}_i=\sum(kc_i)\mathbf{v}_i$; both are again combinations of $\mathbf{v}_1,\dots,\mathbf{v}_p$.

□

**Worked example: a set given by parameters.** $H=\{(s+t,\ s,\ 2t) : s,t \text{ real}\}$. Splitting by parameter, $(s+t,s,2t)=s(1,1,0)+t(1,0,2)$, so $H=\mathrm{Span}\{(1,1,0),(1,0,2)\}$, a subspace of $\mathbb{R}^3$ by Theorem 1.

**The question answered.** $H=\{p\in\mathbb{P}_2 : p(1)=0\}$ is a subspace of $\mathbb{P}_2$: the zero polynomial has value 0 at 1; if $p(1)=q(1)=0$ then $(p+q)(1)=p(1)+q(1)=0$; and $(cp)(1)=c\,p(1)=0$. The set $\{p\in\mathbb{P}_2 : p(1)=1\}$ is not, since the zero polynomial has value $0\neq 1$ at 1. Span and linear combination make sense in any set closed under sums and multiples, and the three subspace conditions test exactly that.

**Shortcuts**

- **Is $H$ a subspace?** First check $\mathbf{0}\in H$; if not, stop. Reason: condition (a). Instance: $\{p : p(1)=1\}$.
- **"All vectors of the form ..."** with expressions linear in the parameters and no constant terms: split by parameter; it is a span, hence a subspace (Theorem 1). Instance: the worked example.
- **A condition preserved by sums and multiples**: check (b) and (c) directly, as in the question.

**Traps**

- The set defined by $x_1+x_2=1$ is not a subspace: it misses $\mathbf{0}$.
- Closed under addition is not enough. The polynomials with integer coefficients contain $\mathbf{0}$ and are closed under addition, but $\frac{1}{2}\cdot 1=\frac{1}{2}$ does not have an integer coefficient, so (c) fails.
- Closed under scalar multiplication is not enough either. The set $\{(a,b) : ab\geq 0\}$ (the first and third quadrants) contains every multiple of each of its vectors, but $(1,0)+(0,-1)=(1,-1)$ is not in it.
- The three conditions must hold for *all* vectors and scalars; one example of a sum that stays inside proves nothing.

## §2.4 Null spaces, column spaces, and linear transformations (Lay 4.2)

**The question.** For the $3 \times 5$ matrix $A$ below, is $(2, 1, 0, 0, 0)$ a solution of $A\mathbf{x} = \mathbf{0}$? Is $(1, 2, 4)$ a combination of the columns of $A$? Is $(1, 2, 3)$? How do we describe all such vectors?

$$A = \begin{bmatrix} 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 2 & -2 \\ 2 & -4 & 1 & 0 & 4 \end{bmatrix}$$

**Definition.** For an $m \times n$ matrix $A$: the **null space** is $\operatorname{Nul} A = \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \mathbf{0}\}$; the **column space** $\operatorname{Col} A$ is the span of the columns of $A$, a subset of $\mathbb{R}^m$; the **row space** $\operatorname{Row} A$ is the span of the rows of $A$, written as vectors in $\mathbb{R}^n$.

The rows of $A$ are the columns of $A^T$, so $\operatorname{Row} A = \operatorname{Col} A^T$. By (W4), $\operatorname{Col} A = \{A\mathbf{x} : \mathbf{x} \in \mathbb{R}^n\}$, and $\operatorname{Col} A = \mathbb{R}^m$ exactly when $A$ has a pivot in every row.

**Theorem** (Null space (Lay 4.2, Thm. 2) and column space (Lay 4.2, Thm. 3))
$\operatorname{Nul} A$ is a subspace of $\mathbb{R}^n$, and $\operatorname{Col} A$ is a subspace of $\mathbb{R}^m$.

*Proof.* 1. $A\mathbf{0} = \mathbf{0}$, so $\mathbf{0}$ is in $\operatorname{Nul} A$.

2. If $A\mathbf{u} = \mathbf{0}$ and $A\mathbf{v} = \mathbf{0}$, then by linearity $A(\mathbf{u} + \mathbf{v}) = A\mathbf{u} + A\mathbf{v} = \mathbf{0}$ and $A(c\mathbf{u}) = c(A\mathbf{u}) = \mathbf{0}$.

3. $\operatorname{Col} A$ is a span, so it is a subspace by Theorem 1.

□

**An explicit description of $\operatorname{Nul} A$.** Solve $A\mathbf{x} = \mathbf{0}$ in parametric vector form; the vectors attached to the free variables span $\operatorname{Nul} A$ (W4). For the question's matrix, $R_3 - 2R_1$ and then $R_3 - R_2$ give the reduced echelon form:

$$\begin{bmatrix} 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 2 & -2 \\ 2 & -4 & 1 & 0 & 4 \end{bmatrix} \xrightarrow{R_3 - 2R_1} \begin{bmatrix} 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 1 & 2 & -2 \end{bmatrix} \xrightarrow{R_3 - R_2} \begin{bmatrix} 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

So $x_1 = 2x_2 + x_4 - 3x_5$ and $x_3 = -2x_4 + 2x_5$, with $x_2, x_4, x_5$ free, and

$$\operatorname{Nul} A = \operatorname{Span}\left\{ \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

**This spanning set is linearly independent.** The vector for the free variable $x_k$ has 1 in entry $k$ and 0 in the entries of the other free variables. So in a combination $c_2\mathbf{u} + c_4\mathbf{v} + c_5\mathbf{w}$, entries $2, 4, 5$ are $c_2, c_4, c_5$, and the combination is $\mathbf{0}$ only if all three weights are 0.

**Writing a subspace as a null space or a column space.**

- **Homogeneous equations give a null space.** $\{(a, b, c) : a + b = 2c,\ b = 3a\}$ is the set of solutions of $a + b - 2c = 0$, $-3a + b = 0$, so it is Nul of the matrix below.

$$\begin{bmatrix} 1 & 1 & -2 \\ -3 & 1 & 0 \end{bmatrix}$$

- **Parameters give a column space.** Put the coefficients of each parameter into one column:

$$\left\{ \begin{bmatrix} 2s - t \\ s + 3t \\ 4t \end{bmatrix} : s, t \text{ real} \right\} = \left\{ s \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 3 \\ 4 \end{bmatrix} \right\} = \operatorname{Col} \begin{bmatrix} 2 & -1 \\ 1 & 3 \\ 0 & 4 \end{bmatrix}.$$

| $\operatorname{Nul} A$ | $\operatorname{Col} A$ |
|---|---|
| a subspace of $\mathbb{R}^n$ (inputs) | a subspace of $\mathbb{R}^m$ (outputs) |
| defined implicitly: the condition $A\mathbf{x} = \mathbf{0}$ | defined explicitly: combinations of the columns |
| finding members takes row reduction | the columns are members; no work |
| $\mathbf{v} \in \operatorname{Nul} A$? Compute $A\mathbf{v}$. | $\mathbf{v} \in \operatorname{Col} A$? Row reduce $[\,A \quad \mathbf{v}\,]$. |
| $\operatorname{Nul} A = \{\mathbf{0}\} \iff A\mathbf{x} = \mathbf{0}$ has only the trivial solution $\iff \mathbf{x} \mapsto A\mathbf{x}$ is one-to-one | $\operatorname{Col} A = \mathbb{R}^m \iff A\mathbf{x} = \mathbf{b}$ is consistent for every $\mathbf{b}$ $\iff \mathbf{x} \mapsto A\mathbf{x}$ is onto $\mathbb{R}^m$ |

**The question answered.** By the row-vector rule,

$$A\begin{bmatrix}2\\1\\0\\0\\0\end{bmatrix} = \begin{bmatrix}2-2\\0\\4-4\end{bmatrix} = \mathbf{0},$$

so $(2,1,0,0,0)$ is in $\operatorname{Nul} A$ (it is the first spanning vector). For the column space, apply the same row operations to $[\,A \quad \mathbf{v}\,]$, first with $\mathbf{v} = (1,2,4)$ and then with $\mathbf{v} = (1,2,3)$:

$$\left[\begin{array}{ccccc|c}1&-2&0&-1&3&1\\0&0&1&2&-2&2\\2&-4&1&0&4&4\end{array}\right] \to \left[\begin{array}{ccccc|c}1&-2&0&-1&3&1\\0&0&1&2&-2&2\\0&0&0&0&0&0\end{array}\right],$$

$$\left[\begin{array}{ccccc|c}1&-2&0&-1&3&1\\0&0&1&2&-2&2\\2&-4&1&0&4&3\end{array}\right] \to \left[\begin{array}{ccccc|c}1&-2&0&-1&3&1\\0&0&1&2&-2&2\\0&0&0&0&0&-1\end{array}\right].$$

So $(1,2,4)$ is in $\operatorname{Col} A$ (it is column 1 plus twice column 3), and $(1,2,3)$ is not (last row $0 = -1$). All solutions of $A\mathbf{x} = \mathbf{0}$ form $\operatorname{Nul} A$, the span of the three vectors above; all combinations of the columns form $\operatorname{Col} A$.

**Linear transformations between vector spaces.** $T\colon V \to W$ is **linear** if $T(\mathbf{u}+\mathbf{v}) = T(\mathbf{u})+T(\mathbf{v})$ and $T(c\mathbf{u}) = cT(\mathbf{u})$ for all $\mathbf{u}, \mathbf{v}$ in $V$ and all scalars $c$. Its **kernel** is $\{\mathbf{u} \in V : T(\mathbf{u}) = \mathbf{0}\}$ and its **range** is $\{T(\mathbf{x}) : \mathbf{x} \in V\}$.

**Theorem** (Kernel and range (Lay 4.2))

The kernel of a linear $T\colon V \to W$ is a subspace of $V$, and its range is a subspace of $W$. For $T(\mathbf{x}) = A\mathbf{x}$ they are $\operatorname{Nul} A$ and $\operatorname{Col} A$.

*Proof.* 1. $T(\mathbf{0}) = T(0 \cdot \mathbf{0}) = 0\,T(\mathbf{0}) = \mathbf{0}$, so $\mathbf{0}$ is in the kernel and in the range.

2. Kernel: if $T(\mathbf{u}) = T(\mathbf{v}) = \mathbf{0}$, then $T(\mathbf{u}+\mathbf{v}) = \mathbf{0}+\mathbf{0} = \mathbf{0}$ and $T(c\mathbf{u}) = c\mathbf{0} = \mathbf{0}$.

3. Range: $T(\mathbf{x}) + T(\mathbf{y}) = T(\mathbf{x}+\mathbf{y})$ and $cT(\mathbf{x}) = T(c\mathbf{x})$ are again images.

4. For $T(\mathbf{x}) = A\mathbf{x}$ the kernel is $\{\mathbf{x} : A\mathbf{x} = \mathbf{0}\} = \operatorname{Nul} A$ and the range is $\{A\mathbf{x}\} = \operatorname{Col} A$.

□

**Worked example: differentiation.** $D\colon \mathbb{P}_3 \to \mathbb{P}_2$, $D(p) = p'$, is linear because $(p+q)' = p'+q'$ and $(cp)' = cp'$.

1. Kernel: for $p = a_0+a_1t+a_2t^2+a_3t^3$, $p' = a_1+2a_2t+3a_3t^2$ is the zero polynomial exactly when $a_1 = a_2 = a_3 = 0$. So the kernel is the constant polynomials.

2. Range: every $b_0 + b_1t + b_2t^2$ is a derivative, namely

$$\frac{d}{dt}\left(b_0t + \frac{b_1}{2}t^2 + \frac{b_2}{3}t^3\right) = b_0 + b_1t + b_2t^2,$$

so the range is all of $\mathbb{P}_2$.

In the same way, $y \mapsto y'' + \omega^2 y$ is linear from the twice differentiable functions to the functions, so the solution set of $y'' + \omega^2 y = 0$, its kernel, is a subspace.

**Shortcuts**

- **Spanning set for** $\operatorname{Nul} A$: one vector per free variable, from the parametric vector form; it is automatically independent. Reason: an explicit description of $\operatorname{Nul} A$, and this spanning set is linearly independent. Instance: the question.
- **Membership tests**: $\mathbf{v} \in \operatorname{Nul} A$? Compute $A\mathbf{v}$. $\mathbf{v} \in \operatorname{Col} A$? Row reduce $[\,A \;\; \mathbf{v}\,]$. Reason: the definitions and (W4). Instance: the question.
- **Recognize the space**: homogeneous equations give Nul; parameters give Col; either way it is a subspace. Instance: the two examples above.
- **Sizes**: for $m \times n$ $A$, $\operatorname{Nul} A \subseteq \mathbb{R}^n$ (number of columns) and $\operatorname{Col} A \subseteq \mathbb{R}^m$ (number of rows). Instance: for the question's $3 \times 5$ matrix, $\operatorname{Nul} A \subseteq \mathbb{R}^5$ and $\operatorname{Col} A \subseteq \mathbb{R}^3$.

**Traps**

- A vector with the wrong number of entries cannot be in the space: $(1, 2)$ is never in $\operatorname{Col} A$ for a matrix $A$ with three rows, whatever its columns.
- The columns of the reduced echelon form do not span $\operatorname{Col} A$ in general. For the matrix below, the RREF has columns $(1, 0)$ and $(2, 0)$, spanning the $x_1$-axis, but $\operatorname{Col} A = \operatorname{Span}\{(1, 2)\}$.

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \xrightarrow{R_2 - 2R_1} \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}$$

- $A\mathbf{x} = \mathbf{b}$ consistent for one $\mathbf{b}$ does not give $\operatorname{Col} A = \mathbb{R}^m$: for the question's matrix $(1, 2, 4)$ is in $\operatorname{Col} A$ but $(1, 2, 3)$ is not.

## §3 Homework this week

HW4, Sections 3.2, 3.3, 4.1, and HW5, Section 4.2.

**Section 3.2.**

- **#2, #3** (which row operation?). Read off the operation, then Theorem 3 (a factor pulled out of a row is (c); a replacement is (a)).
- **#5, #9** (determinant by reduction). Replacements and interchanges only; $\det = (-1)^r \cdot$(diagonal product).
- **#15, #19, #20** (from a $3 \times 3$ matrix with known determinant). Name the operation that produced each matrix and apply Theorem 3.
- **#23, #24** (invertible? independent?). Theorem 4; for vectors, put them in as columns.
- **#27, #29, #31** (true/false). Theorem 3(a); Theorem 4 with dependent columns; three interchanges give $(-1)^3$.

**Section 3.3.**

- **#3, #5** (Cramer's rule). One determinant $A_i(\mathbf{b})$ for each unknown.
- **#9** (Cramer's rule with $s$). $\det A$ as a polynomial in $s$; unique exactly when it is nonzero (worked example).
- **#11** (inverse by the adjugate). Nine cofactors, then transpose (worked example).
- **#21** (parallelogram from vertices). Move one vertex to $\mathbf{0}$; the two adjacent sides are the columns.
- **#27** (area of an image). Theorem 10: multiply by $|\det A|$.
- **#29, #30** (triangle areas). Half a parallelogram; the triangle shortcut.

**Section 4.1.**

- **#2, #3, #6, #7** (subspace?). Find the condition that fails: $\mathbf{0}$, sums, or multiples (traps of 4.1).
- **#10, #11** (sets given by parameters). Write the set as a span; Theorem 1.
- **#20** (functions on $[a, b]$). The zero function, and sums and multiples of continuous functions with $f(a) = f(b)$.

- **#23, #28, #32** (true/false). The zero vector of a function space is the zero function; $V$ is a subspace of itself; the conditions must hold for all vectors.

**Section 4.2.**

- **#1** (is $\mathbf{w}$ in $\operatorname{Nul} A$?). Compute $A\mathbf{w}$.
- **#5** (spanning set for $\operatorname{Nul} A$). One vector per free variable.
- **#9, #13, #15** (write a set as Nul or Col). Homogeneous equations give a matrix whose null space it is; parameters give the columns.
- **#17, #24** (which $\mathbb{R}^k$; membership). Sizes shortcut; compute $A\mathbf{w}$, or row reduce $[\,A \quad \mathbf{w}\,]$.
- **#25, #31, #32, #36, #38** (true/false). Definitions; consistency for *every* $\mathbf{b}$; linearity of differentiation; $\operatorname{Row} A = \operatorname{Col} A^T$.
- **#40** (a second solution). If $A\mathbf{x} = \mathbf{b}$ then $A(5\mathbf{x}) = 5\mathbf{b}$.
- **#43, #46** (linear maps on polynomials and functions). Check the two rules; find the kernel from the defining condition, as in the differentiation example.

## §4 Problem set

**P1.** Compute the determinant below by row reduction.

$$\det \begin{bmatrix} 2 & 4 & -2 & 6 \\ 1 & 3 & 0 & 2 \\ -1 & -1 & 4 & 1 \\ 3 & 8 & 1 & 5 \end{bmatrix}$$

**P2.** $A$ and $B$ are $3\times 3$ with $\det A = 3$ and $\det B = -2$. Find $\det(AB)$, $\det(A^{-1})$, $\det(A^T)$, $\det(2A)$ and $\det(B^T A^{-1} B)$.

**P3.** For which $h$ is the matrix below invertible?

$$\begin{bmatrix} 1 & 2 & h \\ 2 & 5 & 1 \\ 0 & 1 & 3 \end{bmatrix}$$

**P4.** Use Cramer's rule to solve $5x_1 + 2x_2 = 1$, $3x_1 + x_2 = 4$.

**P5.** Find the area of the parallelogram with vertices $(-1, 0)$, $(0, 5)$, $(1, -4)$, $(2, 1)$. Then find the area of its image under the map below.

$$T(\mathbf{x}) = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \mathbf{x}$$

**P6.** Which are subspaces? (a) $\{(a, b, c) \in \mathbb{R}^3 : a + b = 0\}$. (b) $\{(a, b) \in \mathbb{R}^2 : b = a^2\}$. (c) $\{p \in \mathbb{P}_2 : p(0) = 1\}$. (d) $\{(s - 2t,\ s,\ 3t) : s, t \text{ real}\}$. (e) The set of $2 \times 2$ matrices $A$ with $A^T = A$.

**P7.** For the matrix below, find a spanning set for $\operatorname{Nul} A$, and decide whether $\mathbf{b} = (3, 3, -4)$ is in $\operatorname{Col} A$.

$$A = \begin{bmatrix} 1 & -3 & -4 \\ -4 & 6 & -2 \\ -3 & 7 & 6 \end{bmatrix}$$

**P8.** For the matrix below, $\mathbf{u} = (3, -2, -1, 0)$ and $\mathbf{v} = (3, -1, 3)$: is $\mathbf{u}$ in $\operatorname{Nul} A$? Is $\mathbf{v}$ in $\operatorname{Col} A$? Could $\mathbf{u}$ be in $\operatorname{Col} A$?

$$A = \begin{bmatrix} 2 & 4 & -2 & 1 \\ -2 & -5 & 7 & 3 \\ 3 & 7 & -8 & 6 \end{bmatrix}$$